

Practice — Non-contact Forces

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 • MCQ**[1]**

Which of these is a **non-contact** force?

- (a) muscular force (b) friction (c) gravitational force (d) a push from a hand

Q2 • MCQ**[1]**

Two like poles of magnets brought close will —

- (a) attract (b) repel (c) do nothing (d) stick together

Q3 • MCQ**[1]**

Gravitational force is —

- (a) always attractive (b) always repulsive (c) sometimes repulsive (d) a contact force

Q4 • MCQ**[1]**

A plastic scale rubbed with polythene picks up bits of paper because of —

- (a) magnetic force (b) electrostatic force (c) muscular force (d) friction

Q5 • ASSERTION-REASON**[1]**

A: Two balloons rubbed with wool repel each other.

R: Like charges repel.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK

[1]

The two kinds of static charge are positive and _____.

Q7 · FILL IN THE BLANK

[1]

The force with which the Earth pulls objects towards itself is called _____.

Short answer (2 marks each)

Q8

[2]

What is a non-contact force? Name three.

Q9

[2]

State the rule for like and unlike charges (and magnetic poles).

Q10

[2]

How does gravitational force differ from magnetic and electrostatic forces?

Long / application (3 marks each)

Q11

[3]

A charged balloon attracts the woollen cloth used to rub it, but two such balloons repel each other. Explain both observations.

Q12

[3]

Classify these forces as contact or non-contact and justify: (i) friction (ii) magnetic force (iii) gravity (iv) muscular force.

[Check yourself](#) → Answer key